

AERIAL CAMERA

Drone Camera Settings for Video

Frame Rate, ND Filters & the Settings for Smooth Aerials

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What's Inside

Frame Rate, ND Filters & the Settings for Smooth Aerials. Written for pilots whose footage looks juddery, over-sharp, or flat.

- 01 Why Aerial Footage Looks Juddery or Flat**

- 02 Frame Rate & the Shutter Rule in the Air**

- 03 ND Filters — The Aerial Must-Have**

- 04 Resolution, Bitrate & Codec**

- 05 Picture Profiles — Normal vs. D-Log/Flat**

- 06 White Balance — Lock It, Don't Let It Drift**

- 07 Exposure Tools — Histogram, Warnings & ISO Discipline**

- 08 Handling Tricky Light — Bright Sky Over Dark Ground**

- 09 Settings Recipes by Scene**

- 10 Your Pre-Flight Camera Checklist**

START HERE

Why Aerial Footage Looks Juddery or Flat

Your drone flies beautifully. It holds position in wind, glides on command, and shows you the world from a place your feet will never stand. So why does the footage look like a bad slideshow — stuttery pans, plasticky over-sharp edges, or washed-out gray skies? The problem is almost never the aircraft. It is the camera settings, and every one of them is something you can fix on the ground before you take off.

Great aerial video comes down to a handful of decisions most pilots never touch, because the drone flies fine on defaults. But "flies fine" and "cinematic" are two different things. A modern drone camera is capable of buttery motion and rich, gradeable color — it just ships set up for snapshots, not for film. This guide walks you through every setting that matters, in the order that matters, with real numbers you can dial in.

THE WHOLE GUIDE IN ONE SENTENCE

Smooth, professional aerial video is a shutter speed locked to twice your frame rate, held there by an ND filter — everything else is refinement.

The three problems this guide solves

Judder

Pans and fast moves **stutter** instead of gliding. Almost always a shutter speed that is too fast — the classic missing-ND mistake.

Over-sharp

Edges look **crunchy and digital**, foliage shimmers. Usually an aggressive default picture profile plus a shutter that is freezing every micro-detail.

Flat

Footage looks **gray and lifeless** straight off the card. That is a flat/D-Log profile doing its job — it needs grading, and this guide explains the tradeoff.



Gear-agnostic on purpose

Everything here applies to DJI-style aerial cameras and their equivalents. Where a feature has a brand-specific name, we describe what it does so you can find it in your own menus. We never give menu tap-paths, because those change with every firmware update — the concepts do not.

The one idea the whole guide is built on

Video is not a series of frozen photos — it is a series of frames that each contain a little bit of motion. That tiny amount of blur inside each frame is what your eye reads as smooth movement. In a still photo, motion blur is the enemy; in video, a small, consistent blur in every frame is what makes 24 or 30 images per second flow together. When each frame is razor-sharp and frozen, the gaps between frames become visible as a stutter — that is judder.

Sharp frames make stuttery video. A little blur in every frame is what makes motion look real.

The chain that controls motion

Three settings work together to control how motion looks. Get them in this order and everything else falls into place:

1 **Frame rate.**

How many frames per second you capture. Sets the pace and the look — cinematic, standard, or slow-motion.

2 **Shutter speed.**

How long each frame is exposed, which sets how much motion blur lives inside it. This is the setting that makes or breaks smoothness.

3 **ND filter.**

A dark filter over the lens so you can keep that correct-but-slow shutter speed even in blazing daylight without blowing out the sky.



The one-line reason ND filters exist

The "correct" video shutter speed is slow — often 1/50s or 1/60s. In bright sky that slow shutter lets in far too much light and overexposes everything. An ND filter is sunglasses for the lens: it cuts the light so you can keep the slow shutter. Without one, you are forced into a fast shutter, and a fast shutter is what makes aerial footage judder.

That is the entire problem in a nutshell. The next two chapters teach the shutter rule and the ND filter that enforces it — the two settings that separate amateur aerals from footage that looks like film.

CHAPTER 2

Frame Rate & the Shutter Rule in the Air

This is the most important chapter in the guide. Master the relationship between frame rate and shutter speed and you have solved judder forever. It is a single rule, it never changes, and it works on every camera ever made.

Pick your frame rate first

Frame rate sets the feel of your footage and decides whether you can slow it down later. Choose it before anything else, because your shutter speed depends on it.

Frame rate	The look	Use it for
24fps	Cinematic, filmic motion	Landscapes, storytelling, that "movie" feel
25fps	Cinematic (PAL regions)	Same as 24 where 25 is the local standard
30fps	Smooth, "video" standard	Real estate, events, social media, broadcast
60fps	Ultra-smooth; slows to 2× in a 30p timeline	Reveals, orbits, anything you may slow down
120fps	Dramatic slow motion (4×–5×)	Hero moments, water, action beats



Match your project, then commit

If your final video is 24fps, shoot your main coverage at 24 and only jump to 60 or 120 for shots you plan to slow down. Mixing frame rates is fine — just decide the base rate of your project first so the slowed clips have somewhere to live.

The 180-degree shutter rule

Here is the rule that fixes judder, borrowed straight from a hundred years of cinema. It is called the 180-degree shutter rule, and it says:

THE RULE THAT FIXES JUDDER

Set your shutter speed to roughly twice your frame rate. At 24fps use 1/48s (or the nearest 1/50s). At 30fps use 1/60s. At 60fps use 1/120s.

That doubling produces exactly the right amount of motion blur per frame — enough to blend the frames into smooth motion, not so much that everything smears. Most cameras have no exact 1/48s, so 1/50s is the accepted stand-in. Round to the nearest available speed and you are done.

The shutter rule at every frame rate

MEMORIZE THIS

24fps → 1/50s

25fps → 1/50s

30fps → 1/60s

60fps → 1/120s

120fps → 1/240s

Shutter = 2 × frame rate, rounded to the nearest available speed. This never changes, in any light, on any camera.

Why breaking the rule looks bad

⚡ Shutter too fast

Set 1/500s or 1/1000s and each frame freezes hard. Pans and fast subjects **strobe and judder** — the tell-tale stutter of drone footage shot without an ND filter. This is the single most common aerial mistake.

🌊 Shutter too slow

Set 1/15s or 1/25s and each frame smears. Motion turns to **mush**, and any vibration in the aircraft becomes visible wobble. Rare in daylight, but it happens at dusk when you forget to remove the ND.

⚠️ Your drone will not enforce this for you

Left on Auto, the camera raises the shutter as high as it needs to in bright light — straight past 1/1000s — because it is protecting the exposure, not the look. You must switch to **manual (or shutter-priority) control and lock the shutter yourself**. An Auto shutter is the number-one cause of juddery aerials.

🎯 See it for yourself

On the ground, shoot a 5-second clip of passing cars at 1/1000s, then the same shot at 1/50s. Play both back. The fast-shutter clip stutters; the slow one glides. Once you have seen the difference on your own screen, you will never fly on Auto shutter again.

CHAPTER 3

ND Filters — The Aerial Must-Have

Here is the problem the shutter rule creates, and the tool that solves it. In bright daylight, a slow shutter like 1/50s lets in a flood of light. Even at your lowest ISO and a normal aperture, the image blows out — the sky turns to white paper, clouds vanish, snow and sand become featureless glare. You physically cannot keep the correct shutter speed in bright light without help.

That help is a **neutral density (ND) filter**: dark, color-neutral glass that clips over the lens and cuts incoming light by a fixed amount, letting you hold 1/50s at high noon. For aerial video it is not an accessory — it is required equipment.

WHY ND IS THE AERIAL MUST-HAVE

An ND filter is the only thing that lets you obey the shutter rule in bright sky. No ND means a fast shutter means judder. This is why every serious drone pilot carries a set.

How ND strength is labeled

ND filters are rated by how much light they block, usually as a number: ND4, ND8, ND16, ND32, ND64. Each doubling of the number blocks one more stop of light — that is, cuts the light in half again. Bigger number, darker filter, brighter conditions.

Filter	Light cut	Best for
ND4	2 stops	Overcast, heavy clouds, deep golden hour
ND8	3 stops	Cloudy-bright, shade, early morning / late afternoon
ND16	4 stops	Hazy sun, partly cloudy, the everyday daylight filter
ND32	5 stops	Bright sun, blue sky, most sunny-day flying
ND64	6 stops	Blazing midday, snow, water glare, desert, beach



How to pick the strength in the field

Set your frame rate and lock the shutter to twice it. Set ISO to base (usually 100). Now look at the exposure: if it is too bright, add a stronger ND until the image sits at a correct exposure *with the shutter still locked*. You are not changing shutter or ISO to fix brightness — you are choosing the ND that lets them stay put. That is the entire job of the filter.

A working rule of thumb by light

Which ND for which sky

STARTING POINTS

Overcast → **ND4-8**

Cloudy-bright → **ND8-16**

Hazy sun → **ND16**

Full sun → **ND32**

Midday glare → **ND64**

These get you within a stop. Fine-tune by watching the exposure with your shutter locked. Carry the whole set — the sun moves, and so should your filter.



Buy the set, not one filter

ND filters are sold in kits (ND8/16/32/64 is the classic bundle) for good reason: no single strength covers a full day of flying. Get a set matched to your drone, keep them in the case, and swap on the ground between flights. Look for filters described as color-neutral — a cheap filter can add a green or magenta cast you will fight in the edit.



Match ND to the light, not the habit

Leaving an ND64 on into golden hour will force your ISO up and add noise; leaving an ND4 on at noon puts you right back into fast-shutter judder. Re-check your filter every time the light changes meaningfully — a passing cloud bank or the sun dropping toward the horizon is your cue.

CHAPTER 4

Resolution, Bitrate & Codec

With motion handled, we turn to image quality — how much detail and color information the camera records. Three settings control it: resolution, bitrate, and codec. Together they decide how far your footage holds up when you grade, crop, or scale it.

Resolution — 4K is the sensible default

Shoot **4K** whenever you can. It gives a sharp master and lets you crop, reframe, stabilize, or punch in while still delivering a clean 1080p final. Even for a 1080p project, downscaled 4K looks noticeably crisper than footage shot at 1080p natively. Reserve higher frame rates (60/120fps) for slow-motion shots, since those often step the resolution down.



The crop-in bonus

Delivering in 1080p but shooting 4K means you can push into a shot by up to 2× with no loss of final sharpness — a free second "camera move" you create in the edit. It also gives stabilization room to work without softening the image.

Bitrate — how much data per second

Bitrate is how many megabits per second the camera writes to the card. Higher bitrate means more detail preserved, cleaner gradients in skies, and fewer compression artifacts (the blocky mush you see in fast water or foliage). If your camera offers a high-bitrate or "H.265" mode, use it — especially for flat profiles, which need every bit of data to grade cleanly.

Quality settings that hold up

RECOMMENDED

Resolution **4K**

Bitrate **Highest available**

Codec **H.265 / HEVC**

Highest bitrate plus H.265 gives you the most gradeable, artifact-free file your card and drone can produce. Bring fast, high-capacity cards to keep up.

Codec — H.264 vs. H.265

H.264

Older, universally compatible, easier on slow editing computers. Fine for quick social clips. Uses more data for the same quality, so gradients can band.

H.265 / HEVC

Newer, more efficient — more quality per megabit, better for flat profiles and rich skies. Needs a reasonably modern computer to edit smoothly. The better choice when your machine can handle it.



The card is part of the camera

High bitrate and high frame rate demand a fast microSD card rated for sustained video writing (look for V30 or higher). A slow or off-brand card can drop frames or stop recording mid-flight. Buy the card the manufacturer recommends, and reformat it in the drone before a shoot.

CHAPTER 5

Picture Profiles — Normal vs. D-Log/Flat

This setting decides how your footage looks straight off the card — and it carries a genuine tradeoff. A picture profile (also called a color profile or color mode) controls how much contrast and saturation the camera "bakes in" before saving. There are two camps, and choosing wrong for your workflow will cost you.

Normal / Standard — ready to use

The normal profile applies contrast and color for you, so footage looks punchy and finished right out of the drone. Skies are blue, greens are green, and you can post it with little or no editing. The tradeoff: those decisions are permanent. Highlights that clipped to white are gone, and there is less room to push the look elsewhere later.

Normal / Standard profile

GRADE-FREE

Look **Finished**

Editing **Minimal**

Latitude **Lower**

Choose this when you want to deliver fast, do not plan to color grade, or are still learning. It looks good with zero work.

D-Log / flat — maximum latitude, requires grading

A flat or log profile (D-Log, D-Log M, and similar) does the opposite: it records the widest range of brightness and the most color information, leaving contrast and saturation for *you* to add in editing. Straight off the card it looks gray, washed out, and lifeless — exactly as intended. All that flatness is preserved shadow and highlight detail, waiting to be shaped.

THE FLAT-PROFILE TRADEOFF

Flat/log footage holds far more detail in bright skies and dark ground — but it looks gray and dead until you grade it. If you will not color grade, do not shoot flat.



Flat is not a shortcut to "cinematic"

Many new pilots switch to D-Log because it looks professional in tutorials, then post the gray, ungraded files and wonder why the footage looks worse than before. Flat footage is an **ingredient, not a dish**. If it goes straight to social media without grading, you have thrown away quality for nothing. Only shoot flat if you will take it into an editor and add contrast and color back.

How to grade a flat clip, briefly

1

Apply the matching conversion LUT.

Manufacturers provide a LUT (a color-conversion preset) that turns their specific log profile back into a natural-looking image. This is your starting point, not the finish line.

2

Set contrast and white/black points.

Bring the grays back to a full range so blacks are black and highlights have life.

3

Adjust saturation and balance to taste.

Now push the mood — cool and crisp, or warm and golden. This is where flat footage pays off.



The honest recommendation

If you are newer, or you need to deliver quickly, shoot the **Normal** profile and enjoy footage that looks great with no work. Move to D-Log only once color grading is a step you actually do. Both are valid — the mistake is shooting flat and then not grading it.

CHAPTER 6

White Balance — Lock It, Don't Let It Drift

White balance controls how warm or cool your footage looks — whether whites read as neutral, orange, or blue. On the ground it is easy to ignore. In the air, on Auto, it becomes a quiet ruin of otherwise good footage.

Why Auto white balance sabotages video

Auto white balance re-evaluates the scene constantly. As your drone flies over a green field, a gray road, then a blue lake, the camera keeps "correcting," and the color temperature **shifts and pulses** across the shot. On a still you would never notice. On moving video the whole frame subtly warms and cools as you fly — a shimmering color drift that screams amateur and is nearly impossible to fix cleanly.

THE WHITE BALANCE RULE

Never fly video on Auto white balance. Set a manual Kelvin value and lock it so the color stays consistent from the first frame to the last.

Set it manually by Kelvin

Switch white balance from Auto to manual and dial in a Kelvin value that matches your light. It does not have to be scientifically perfect — it has to be *consistent*. A locked value that is slightly warm is far better than an Auto value that wanders.

White balance by light

STARTING POINTS

Sunny day → **5500K**

Cloudy / shade → **6500K**

Golden hour → **5000-5500K**

Blue hour → **4000K**

Set once for the conditions, then leave it. If you shoot flat/log, get it close — grading can nudge it, but a locked start makes everything easier.



When in doubt, warmer is friendlier

A slightly warm white balance (higher Kelvin) tends to flatter landscapes and skin, and reads as inviting. A slightly cool setting reads as clinical. If you are unsure, err a touch warm — and above all, keep it locked for the whole flight.

CHAPTER 7

Exposure Tools — Histogram, Warnings & ISO Discipline

You cannot trust a small, sun-washed phone or controller screen to judge exposure. In bright daylight the image on your screen looks fine while the footage is actually blowing out the sky. Instead of trusting your eyes, learn to trust the camera's exposure tools — they never lie.

Read the histogram

The histogram is a graph of brightness across the frame: dark tones on the left, bright tones on the right. It tells you the truth your screen cannot. If the graph is jammed hard against the right edge, you are clipping highlights — the sky is turning to featureless white and that detail is gone forever. If it is crushed against the left, your shadows are going pure black. You want the important information sitting inside the graph, not spilling off either end.



Protect the highlights

In aerial video the sky is usually the brightest thing and the easiest to lose. When in doubt, expose so the histogram stays off the right wall — slightly darker footage can be lifted in the edit, but a blown-white sky can never be recovered. Protect the bright end.

Turn on the overexposure warning

Most drones offer an overexposure warning (often called "zebras" or a highlight alert) that paints diagonal stripes over any part of the image that is clipping to white. Turn it on. Now you get an instant, unmistakable signal the moment your sky or a snowfield is blowing out — no guessing on a glary screen. If the zebras appear over important areas, you are overexposed: reach for a stronger ND filter first, then lower ISO, and only touch the shutter as a last resort.



Fix exposure in the right order

When the image is too bright with your shutter correctly locked, adjust in this order: **1) stronger ND filter, 2) lower ISO toward base, 3) close the aperture if your drone has one.** The shutter speed is the last thing you change, because changing it breaks the motion rule from Chapter 2.

ISO discipline

ISO is your sensor's sensitivity to light. Keep it as low as possible — usually the base of 100 — for the cleanest, least noisy image. Because your shutter is locked by the motion rule, you do *not* raise ISO to fix daylight brightness (that is the ND filter's job). You only raise ISO when the light genuinely runs out — deep dusk, blue hour, night — and even then you raise it as little as you can get away with.

ISO by condition

KEEP IT LOW

Daylight → ISO 100

Golden hour → ISO 100-200

Blue hour → ISO 400-800

Night → ISO 800-1600

In daylight you should almost never leave base ISO — if the image is too bright, change the ND, not the ISO. Noise from a high ISO is far harder to fix than a slightly dark exposure.



The pro's daylight loop

Lock frame rate. Lock shutter to twice it. Drop ISO to base. Now choose the ND filter that lands the histogram correctly with zebras clear. Four decisions, same order every flight — and every one of your clips will match in motion, exposure, and feel.

CHAPTER 8

Handling Tricky Light — Bright Sky Over Dark Ground

The hardest situation in aerial video is the most common one: a brilliant sky above dark ground — a forest at sunrise, a city at dusk, mountains against a blazing horizon. The brightness gap between the sky and the ground is larger than the sensor can hold in a single frame, so something has to give. This chapter is about giving up the right thing.

You usually cannot keep both — so choose

When the scene's brightness range exceeds what the camera can capture (its dynamic range), you must decide what matters more: detail in the bright sky, or detail in the dark ground. Expose for both at once and you will clip the sky to white *and* crush the ground to black.



Protect the sky

Expose so the sky keeps its clouds and color, letting the ground go darker. Usually the better choice — a rich sky reads as intentional and cinematic, and shadow detail can be lifted in the grade.



Protect the ground

Expose for the land when the ground *is* the subject, accepting a brighter, blown sky. Works best when the sky is a small part of the frame or overcast anyway.



Compose the gap away

The easiest fix for a brightness gap is to include less of it. Tilt the gimbal down to fill the frame with the ground, or point more skyward to feature the clouds. A top-down or steep-down angle sidesteps the sky-versus-ground problem entirely — no sky in frame, no clipping to fight.

Let the flat profile earn its keep

This is where D-Log pays off. A flat/log profile captures a wider brightness range, so it holds onto both sky detail and shadow detail that a normal profile would clip. If you are flying difficult high-contrast light and you are willing to grade, a flat profile plus careful exposure gives you the most room to balance sky and ground in the edit. This is the single best reason to learn flat-profile grading.



Shoot into the golden hours instead

The reason sunrise and sunset are the aerial pilot's favorite light is not just the color — it is that the sky-to-ground brightness gap shrinks dramatically. Softer, lower light means the sensor can hold the whole scene at once, and tricky-light problems mostly disappear. When you can choose your flight time, choose the hour after sunrise or before sunset.



Never point the lens at the sun for long

Aiming directly into the sun, especially with a strong ND, can flare the image and risks the sensor. Frame the sun at the edge or just out of frame for that glow without the damage, and keep such shots brief.

CHAPTER 9

Settings Recipes by Scene

Here are complete, ready-to-fly setups for the three situations you will meet most. Each is a full recipe — frame rate, shutter, ISO, ND, and white balance — that obeys every rule in this guide. Dial them in, then fine-tune the ND to your exact sky.

Sunny landscape — blue sky, bright sun

Bright sunny landscape

FULL SUN

Frame **24fps**

Shutter **1/50s**

ISO **100**

ND **ND32**

WB **5500K**

The classic cinematic aerial. If the histogram still clips the sky, step up to ND64. Shoot 4K, highest bitrate, and pick Normal or D-Log by whether you will grade.

Golden hour — warm, low, dramatic light

Golden hour glide

SUNRISE / SUNSET

Frame **24fps**

Shutter **1/50s**

ISO **100-200**

ND **ND8-16**

WB **5000-5500K**

Lighter light needs a lighter filter. Keep white balance locked warm to preserve the gold — do not let Auto WB cool it into gray. As the sun drops, be ready to swap to a weaker ND or remove it.

Real estate / overcast — even, soft light

Real estate & overcast

SMOOTH & EVEN

Frame **30fps**

Shutter **1/60s**

ISO **100**

ND **ND4-8**

WB **6000-6500K**

30fps reads as clean, professional "video" for property and events. Overcast is soft, so a lighter ND holds the shutter. A warmer WB keeps a gray day from looking cold and drab.

Fly slow, and it all looks better

Whatever the recipe, slow your movements down. Gentle stick inputs, wide slow orbits, and smooth reveals let your correctly-blurred frames glide. Aggressive, jerky flying makes even perfect settings look frantic — smooth footage is 50% settings and 50% a light touch on the sticks.

Build your own recipe card

Take a screenshot of the recipe that matches your typical light and keep it on your phone. On location, you will dial in a matched, judder-free look in under a minute instead of fiddling in the field. Update it as you learn what your specific drone and skies want.

CHAPTER 10

Your Pre-Flight Camera Checklist

Everything in this guide comes down to a short list of decisions you make on the ground, before you launch. Run this checklist before every flight and your footage will be smooth, correctly exposed, and gradeable — every single time. Tape it inside your case.

Set on the ground, before takeoff

- ☒ Frame rate chosen for the project (24/30 for base, 60/120 for slow motion).
- ☒ Camera in manual or shutter-priority — **not** full Auto.
- ☒ Shutter locked to roughly twice the frame rate (1/50s at 24fps, 1/60s at 30fps).
- ☒ ND filter matched to the light and actually mounted on the lens.
- ☒ ISO at base (100) for daylight; raised only if the light is truly gone.
- ☒ White balance set to a manual Kelvin value and locked — never Auto.
- ☒ Resolution at 4K, highest bitrate, H.265 if your workflow supports it.
- ☒ Picture profile chosen: Normal if you will not grade, D-Log if you will.
- ☒ Overexposure warning (zebras) turned on; histogram visible.
- ☒ Fast, formatted memory card with room to spare.

Confirm in the air, before the real take

- ☒ Histogram sitting inside the graph, not clipping the right edge.
- ☒ No zebras painting over the sky or other important areas.
- ☒ Motion looks smooth on a slow test pan — no strobing or judder.
- ☒ Color looks consistent as you fly over different ground.
- ☒ Gimbal level, and your stick movements slow and deliberate.



Re-check when the light changes

The sun moves, clouds roll through, and golden hour arrives fast. Any real change in the light is your cue to land, swap the ND, and re-confirm exposure. Two minutes on the ground beats a whole card of over- or under-exposed clips.

The whole system, one more time

1

Frame rate sets the pace.

Pick it first; it decides your shutter.

2

Shutter at twice the frame rate kills judder.

Lock it and never let Auto touch it.

3

ND filter holds that shutter in bright light.

The aerial must-have; match it to the sky.

4

ISO stays at base; white balance stays locked.

Clean image, consistent color.

5

Expose to the histogram, protect the sky, grade if you shot flat.

The finish.



Your first flight with this guide

Pick one recipe from Chapter 9, run the pre-flight checklist, and fly a single slow, wide orbit of something you love. Bring it home, play it full-screen, and watch it glide. That first smooth, rich clip is the moment all of this clicks — and you will never fly on defaults again.

REMEMBER

An ND filter holding your shutter at twice the frame rate is the whole secret to smooth aerals — everything else just makes a smooth shot look richer.

Story Mode Guy

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